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# Temperature interactions with transpiration response to vapor pressure deficit among cultivated and wild soybean genotypes

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A key strategy in soybean drought research is increased stomatal sensitivity to high vapor pressure deficit (VPD), which contributes to the 'slow wilting' trait observed in the field. These experiments examined whether temperature of the growth environment affected the ability of plants to respond to VPD, and thus control transpiration rate (TR). Two soybean [*Glycine max* (L.) Merr.] and four wild soybean [*Glycine soja* (Sieb. and Zucc.)] genotypes were studied. The TR was measured over a range of VPD when plants were growing at 25 or 30°C, and again after an abrupt increase of 5°C. In *G. max*, a restriction of TR became evident as VPD increased above 2.0 kPa when temperature was near its growth optimum of 30°C. 'Slow wilting' genotype plant introduction (PI) 416937 exhibited greater TR control at high VPD compared with Hutcheson, and only PI 416937 restrained TR after the shift to 35°C. Three of the four *G. soja* genotypes exhibited control over TR with increasing VPD when grown at 25°C, which is near their estimated growth optimum. The TR control became engaged at lower VPD than in *G. max* and was retained to differing degrees after a shift to 30°C. The TR control systems in *G. max* and *G. soja* clearly were temperature-sensitive and kinetically definable, and more restrictive in the 'slow wilting' soybean genotype. For the favorable TR control traits observed in *G. soja* to be useful for soybean breeding in warmer climates, the regulatory linkage with lower temperatures must be uncoupled.

## Introduction

Crop biomass increases as a linear function of water use (DeWit 1958). The physical relationship between carbon assimilation and transpiration is controlled by gas exchange through the stomata in plant leaves. When stomata are open, $CO_2$ is taken in by the plant and water vapor escapes from the leaf. When water deficits occur and stomata close, the water status and many physiological functions of the plant are maintained, but carbon fixation and growth rate decline.

Drought is a problem in all major crops, and certainly in soybean (Purcell and Specht 2004). Multiple strategies are available to soybean growers to protect crop yield in times of drought. Although irrigation can be effective, it is not widespread because costs commonly exceed the revenue gained from the yield increase. Crop phenology can be matched to seasonal patterns, where sowing

*Abbreviations* – AQP, aquaporin; MG, maturity group; PI, plant introduction; PPFD, photosynthetic photon flux density; RH, relative humidity; $S_1$, the slope at VPD lower than the breakpoint (or, the only slope for the linear model); $S_2$, the slope at VPD greater than the breakpoint; TR, transpiration rate; VPD, vapor pressure deficit.

occurs early in the season when more water is available (e.g. early soybean production systems; Bowers 1995, Heatherly 1999). However, the approach is not feasible in every soybean growing region. An option that is cost-effective and available to most soybean farmers is selection of drought-resistant varieties.

Considerable progress has been made in the breeding of improved drought resistance in soybean based on a 'slow-wilting' (drought resistant) phenotype that was initially observed in the plant introduction (PI) 416937 (USDA-ARS National Genetic Resources Program, 1977) more than 20 years ago (Sloane et al. 1990, Carter and Rufty 1993). Since then, breeding lines developed from this PI have showed relatively high yields in drought conditions without an obvious yield penalty in environments where water is plentiful (Gillen and Shelton 2006, 2007, 2008). One of the key mechanisms of PI 416937 appears to be a constraint of transpiration rate (TR) to a maximum rate at high atmospheric vapor pressure deficit (VPD), with transpiration being restricted at VPD greater than approximately 2.1 kPa (Fletcher et al. 2007). By limiting transpiration to a maximum rate at high VPD, damaging physiological effects of water-deficit stress in plant tissues are delayed and available soil water is conserved.

Regulation of maximum TR and other drought resistance traits have been evaluated using a mechanistic simulation model for their potential effectiveness in improving yields for soybean across the major growing regions in the USA based on 50 years of weather data (Sinclair et al. 2010). Repeated simulations indicated that lowered maximum TR was one of the more promising traits, especially in the southern USA. The South experiences higher temperatures than many areas where soybean is grown, however, and temperature interactions with the trait were not addressed in the model. If expression of the maximum transpiration trait were affected by temperature, yield benefits predicted by simulation models would be inaccurate.

The genetic base for modern-day soybean cultivars in the USA is relatively narrow and cultivars tend to have uniform drought responses. These observations have led to the pursuit of drought resistant traits in diverse germplasm (Carter et al. 1993, 1999, 2004). At least 10 soybean plant introductions are now being used in soybean breeding (T.E. Carter, Jr personal communication). The wild ancestor of soybean, *Glycine soja* (Sieb. and Zucc.), also known as wild soybean, is perhaps the most genetically diverse resource available to soybean breeders that will cross freely with the domesticate (Harlan and de Wet 1971). An annual plant, it is commonly found throughout China and most of Asia, including its arid regions (Hymowitz and Singh 1987). *G. soja* has been the object of two previous water deficit studies; drought resistance was found to be superior to *Glycine max* in one case (Chen et al. 2006) but similar to *G. max* in the other (James et al. 2008). Physiological processes of *G. soja and G. max* were not compared, so the basis for the putative drought resistance is unresolved. There are no reports of using *G. soja* as a source of drought resistance in breeding, in large part because no *G. soja* accessions have been verified as resistant.

In this manuscript, we describe a series of experiments initiated to investigate the transpiration responses to VPD and their interactions with temperature. The experiments used a whole plant chamber system that allowed stepwise increases in VPD and monitoring of TR (Fletcher et al. 2007, Sinclair et al. 2008, Sadok and Sinclair 2009, Devi et al. 2010, Gholipoor et al. 2010, Yang et al. 2012). The first objective was to determine whether increasing temperature modified the VPD response of two genotypes of *G. max*. The experiments were conducted with the cultivar Hutcheson (Buss et al. 1988) and PI 416937. Hutcheson has been widely grown in the Southeast and is among the top 10 cultivars most frequently used as parental stock in soybean breeding in the USA over the past 20 years (Mikel et al. 2010). Its responses were compared with those of the slow wilting PI 416937 (Sloane et al. 1990, Carter and Rufty 1993, Carter et al. 2006). Then, we examined four *G. soja* genotypes to determine if the pattern of their VPD responses or the temperature sensitivities differed from those of soybean. This allowed assessment of whether *G. soja* might become a new source of drought resistance traits for soybean breeding.

## Materials and methods

### Plant growth and genetic material

A sandy loam soil (73% sand, 17% silt and 10% clay) was sterilized and mixed with slow-release fertilizer (19%N, 6%P and 12%K) plus lime to obtain a homogeneous soil media with adequate nutrients and pH. The soil was placed into custom-built PVC pots measuring 32 cm in height and 10 cm in diameter (2.5 l volume). Before being filled with soil, porous ceramic cups were inserted in the bottom of each pot and covered by diatomaceous earth (approximately 250 ml). A vacuum line was connected to the cup within the base of the pot to remove water from the soil and prevent anaerobic conditions. After each water addition during the experiment, the bottoms of the pots were subjected to a vacuum of ≈33 kPa (1/3 bar) until water pulled from the soil was negligible.



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**Table 1.** Description of drought resistance characteristics of wild and domesticated soybean genotypes used in experiments. WUE, water use efficiency.

| Species | Genotype | MG | Geographic adaptation or origin | Description | Reference |
|---|---|---|---|---|---|
| *Glycine max* | Hutcheson | V | Southern USA | Intermediate wilting in field | King et al. (2009) |
| | | | | Intermediate WUE | Hufstetler et al. (2007) |
| | PI 416937 | V | Japan | Slow wilting in field | Sloane et al. (1990); King et al. (2009) |
| | | | | VPD-sensitive transpiration rate | Fletcher et al. (2007) |
| *Glycine soja* | PI 468917 | III | Liaoning, China | Variable wilting with soil drying (pot) | Preliminary screen |
| | PI 597454A | III | Shanxi, China | Variable wilting with soil drying (pot) | Preliminary screen |
| | PI 468397A | III | Shanxi, China | Rapid wilting with soil drying (pot) | Preliminary screen |
| | PI 549033 | IV | Liaoning, China | Slow wilting with soil drying (pot) | Preliminary screen |

The water content of the pots never fell to a level that would induce stress responses in the plants.

Two *G. max* and four *G. soja* genotypes were selected for the study based on results from previous drought studies (Table 1 and references therein). Hutcheson and PI 416937 are both maturity group (MG) V genotypes of domesticated soybean, *G. max*. Hutcheson is a USA cultivar that has intermediate wilting and intermediate water use efficiency, while PI 416937 is a PI from Japan that has slow wilting and VPD-sensitive TR response. Wild soybean, *G. soja*, genotypes (MG III & IV) were selected based on contrasting wilting characteristics in a preliminary greenhouse screen. Soybean seeds were germinated in paper and kept moist by capillary action with a calcium sulphate (100 μM) solution. Seeds were held in a dark germination chamber at 30°C for 48 h. *G. soja* accessions have a hard seed coat and were, thus, scarified prior to germination. Germination procedures were similar to those used for *G. max* except that germination was extended to 72 h. Each pot was planted with two germinated seedlings of *G. max* or with four seedlings of *G. soja* and then thinned to one plant per pot after 7 days.

After planting, pots were placed in two walk-in growth chambers at the Southeastern Plant Environment Laboratory at North Carolina State University (Saravitz et al. 2009). Temperature was held constant at either 25 or 30°C during the growth period, depending on the treatment as described below. The chambers were set for 10-h day-lengths with a combination of high output fluorescent and 100-watt incandescent bulbs providing a photosynthetic photon flux density (PPFD) of approximately 800 μmol $m^{-2}$ $s^{-1}$. The dark period was interrupted by 3 h of incandescent light (PPFD 40 μmol $m^{-2}$ $s^{-1}$) to repress flowering.

## VPD treatments and transpiration

Transpiration was measured as incremental VPD treatments were imposed on plants. Experiments were conducted using a VPD system similar to that widely used and previously described in the literature (Fletcher et al. 2007, Sinclair et al. 2008, Sadok and Sinclair 2009, Devi et al. 2010, Gholipoor et al. 2010). Clear plastic containers (28 cm diameter × 30 cm height; 20-l volume) served as VPD chambers for individual plants. Flanges on the PVC pots secured the VPD chambers in place only during VPD treatment/measurement periods. The system, which was inside a walk-in growth chamber, accommodated 12 VPD chambers in total. Quick connect fittings were installed in the tops of chambers so that air tubes could be inserted and removed easily. A small computer fan (76 mm diameter; Northern Tool and Equipment, Burnsville, MN) was installed inside each chamber to keep the air mixed. Air temperature (T) and relative humidity (RH) were measured once per minute by data loggers with built-in sensors (Lascar Electronics, Whiteparish, UK) that were inserted into the sides of the chambers. VPD (kPa) was calculated from measured T (°C) and RH (%) using the equation $VPD = (1 - RH/100) \times (0.611 + 0.0442 \times T + 0.00146 \times T^2 + 2.32E-5 \times T^3 + 4.56E-7 \times T^4)$. A previous experiment using similar chambers found that leaf and air temperature were similar, so air temperature was suitable for use in the VPD calculation (Sinclair et al. 2008).

Vapor pressure deficit treatments began for soybean when the plant canopy was approximately 25 cm in diameter. At this size, the plants fit inside the chambers without making contact with the sides of the chamber. Because of its prostrate growth habit, *G. soja* plants were trained up a 30-cm high stake inserted in the middle of the pot. The canopy structure of *G. soja* was not as defined as *G. max*. Therefore, VPD treatments were started when leaf area was similar for *G. soja* and *G. max*. Leaf area estimates for *G. soja* were made by destructive harvest of additional plants grown for this purpose. Genotypes within each species reached the desired size at the same time, but between the two



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species, *G. soja* took approximately 1–2 weeks longer than *G. max* to reach a similar leaf area.

Plants were exposed to VPD treatments over two consecutive days and were well watered each evening prior to the next day's measurement. On each day of measurement, three 90-min VPD treatments were imposed on the plants. The first 90-min VPD treatment was targeted for a range of 0.5–1.5 kPa. The second and third treatments were targeted for VPD conditions of 1.5–2.5 and 2.5–3.5 kPa. Plants were given approximately 30 min between treatments to attain a steady state in the new conditions.

VPD inside each VPD chamber was controlled by changing the rate of air flow through the chamber and/or the dryness of the air flowing into the chamber. Typically, the airflow was 3–5 l $min^{-1}$ in the first treatment, 10–15 l $min^{-1}$ for the second and 15–20 l $min^{-1}$ for the third. Also, in the second and third treatments, the air was dried by passing through a PVC tube of 5-cm diameter and 115-cm length filled with silica gel desiccant (Fisher Chemical, Fair Lawn, NJ) before flowing into each of the chambers. Air flowed out of the chambers through small holes in the bottom of the container and a gap between the VPD chamber and the pot. During measurements, temperature was relatively constant, being controlled by the walk-in growth chamber. Temperatures were changed as an experimental treatment on different days and will be discussed later in more detail.

Each pot-plant-VPD chamber unit was detached from air lines for weighing before and after each VPD treatment. The difference in weight (generally in the 5–15 g range) was used as an estimate of transpiration. Evaporation from the soil surface was blocked by aluminum foil. After the last VPD treatment of the second day, the shoots of measured plants were harvested and leaf area measured (LI-3100 Area Meter, LI-COR Inc., Lincoln, NE). VPD was calculated for each minute of each 90-min measurement period using the temperature and relative humidity values recorded inside the chambers by the data loggers, and the VPD was well-controlled during each 90-min period (standard error of the mean <0.02 kPa). TR was expressed per unit of leaf area. Experiments were repeated two to three times for *G. max and G. soja* genotypes and sometimes with multiple runs within an experiment (Table 2). The number of replicate plants varied because of the different number of genotypes investigated for each species and the limited space with the system, but the range was three to six replications per genotype per experiment. For example, Experiment 1 measured

**Table 2.** Summary of experiments measuring TR response to VPD at 25 and 30°C growth temperatures. Includes the VPD measurement temperatures, range of VPD during measurements, and mean leaf area (±SE) of *Glycine max* and *Glycine soja* plants. [a]Mean leaf area not measured because the plants were needed for the following run.

| Growth temperature (°C) | Species | Experiment | Run | Measurement temperature (°C) | VPD range (kPa) | Mean leaf area ± SE ($cm^2$) | Days after planting | Replications |
|---|---|---|---|---|---|---|---|---|
| 25 | *Glycine max* | 1 | a | 25 | 1.19–2.56 | – [a] | | 4 |
| | | | b | 30 | 1.25–3.30 | 174.6 ± 13.9 | 22 | 4 |
| | | | c | 25 | 0.93–2.81 | – | | 4 |
| | | | d | 30 | 1.29–3.50 | 191.5 ± 11.5 | 24 | 4 |
| | | 2 | a | 25 | 0.96–2.87 | – | | 6 |
| | | | b | 30 | 1.36–3.72 | 226.5 ± 21.9 | 26 | 6 |
| | *Glycine soja* | 3 | a | 25 | 0.75–2.34 | – | | 3 |
| | | | b | 30 | 0.89–3.18 | 359.7 ± 42.3 | 38 | 3 |
| | | | c | 25 | 0.62–2.61 | – | | 3 |
| | | | d | 30 | 1.41–3.27 | 388.2 ± 24.6 | 40 | 3 |
| | | 4 | a | 25 | 0.53–2.02 | – | | 3 |
| | | | b | 30 | 0.75–3.01 | 564.7 ± 41.2 | 40 | 3 |
| | | | c | 25 | 0.56–2.39 | – | | 3 |
| | | | d | 30 | 0.72–3.20 | 748.5 ± 35.7 | 42 | 3 |
| 30 | *G. max* | 5 | a | 30 | 0.86–3.00 | – | | 3 |
| | | | b | 35 | 1.43–3.72 | 473.7 ± 45.4 | 19 | 3 |
| | | 6 | a | 30 | 1.18–3.41 | – | | 6 |
| | | | b | 35 | 1.29–3.51 | 294.0 ± 13.2 | 20 | 6 |
| | | 7 | a | 30 | 1.00–3.08 | – | | 6 |
| | | | b | 35 | 1.26–3.55 | 468.1 ± 29.1 | 19 | 6 |
| | *G. soja* | 8 | a | 30 | 1.06–3.30 | – | | 6 |
| | | | b | 35 | 1.27–3.57 | 374.0 ± 24.2 | 22 | 6 |
| | | 9 | a | 30 | 1.14–3.61 | – | | 6 |
| | | | b | 35 | 1.12–3.47 | 518.1 ± 27.4 | 29 | 6 |



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four plants (replications) of each *G. max* genotype on each day. The same plants were measured at three VPD conditions in one day, called Run A, and at three VPD conditions on a second day, called Run B, then those plants were destructively harvested. Then another set of plants that had grown at the same conditions were similarly measured in Run C and Run D, and destructively harvested.

### Temperature treatments

In summer months in the southeastern USA, high VPD commonly coincides with high temperatures. Average daily maximum temperatures range between 30 and 32°C (Sermons et al. 2008, Purcell et al. 2003) and occasionally reach or exceed 35°C. During a day, and sometimes from day to day, plants will experience rapid increase in temperature. TR across a range of VPD was measured during exposure to two types of conditions: (1) a stable temperature, the one at which plants were growing (25 or 30°C) and (2) following an abrupt upward 5°C shift in temperature, imposed in the morning of the measurement day (about 30 min before the first measurement period of the day and 270 min before the last measurement period).

### Experimental design and data analysis

*G. max and G. soja* genotypes were arranged in a completely randomized design for the experiments. Because of space limitations, *G. max and G. soja* genotypes were run in nine separate experiments (Table 2). For a given run within each experiment, consisting of three sequential measurement periods in one day, the TR of each plant was plotted against the mean VPD for each respective 90-min VPD treatment. A graphing and statistical software package, GRAPHPAD PRISM (GraphPad Software, San Diego, CA, 1996), was used to analyze TR response to VPD. First, two types of regressions, a linear regression (Eqn 1) and a two-segment, non-linear regression (Eqn 2), were fit to the data.

$$\mathrm{TR} = S_1 \ (\mathrm{VPD}) + \mathrm{intercept} \quad (1)$$

$$\begin{aligned} &\text{If VPD} \leq X_0, \ \mathrm{TR} = S_1 \ (\mathrm{VPD}) + \mathrm{intercept\ 1} \\ &\text{If VPD} \geq X_0, \ \mathrm{TR} = S_2 \ (\mathrm{VPD}) + \mathrm{intercept\ 2} \end{aligned} \quad (2)$$

In these equations, TR = transpiration rate, $X_0$ = the breakpoint VPD where the slope of TR changes, $S_1$ = the slope at VPD lower than the breakpoint (or, the only slope for the linear model) and $S_2$ = the slope at VPD greater than the breakpoint. After both regressions were fit to each data set, they were compared using F-tests. For each data set that was fit significantly by the two-segment equation ($P < 0.05$), the two-segment model is shown in the figure panel and the *P*-value is given below the $r^2$ (Figs 1–4). When the two-segment model was not significant ($P > 0.05$), the linear equation was accepted, so the linear model is shown in the figure panel.

## Results

### Experimental conditions

All VPD and temperature treatments were successfully carried out within 2°C of their designated temperatures. Prior to the onset of VPD treatments, plants were acclimated to mean daytime VPD values of approximately 1.2 kPa for all experiments where plants were grown at 25°C. Plants growing at the 30°C temperature were exposed to mean daytime VPD values ranging from 1.7 to 2.1 kPa. The maximum VPD achieved during VPD treatments at 25°C was, on average, 2.8 kPa, somewhat lower than the 3.3 and 3.6 kPa at 30 and 35°C (Table 2). This was because of the inherent relationship between saturated vapor pressure and temperature; though a similar minimum relative humidity was achieved by the experimental system at all temperatures, the VPD differed. For example, 25% relative humidity is equivalent to a VPD of 2.4 kPa at 25°C, 3.2 kPa at 30°C and 4.2 kPa at 35°C; 75% relative humidity corresponds to 0.79, 1.1 and 1.4 kPa at the three temperatures.

### *Glycine max*

At a constant temperature of 25°C, Hutcheson's TR increased as a linear function of VPD (Fig. 1A, Table 3). There was no evidence of a limited maximum TR. In contrast, when grown and measured at a constant 30°C, Hutcheson's TR increased linearly at lower VPD, but with progressive increase of VPD 1.93 kPa, TR increased at a much slower rate (Fig. 1B). The slope of the regression before the breakpoint ($S_1$) was 34.0 mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$, and it decreased to 5.85 mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ after the breakpoint ($S_2$) (Table 4). This represented a significant leveling off of TR at high VPD.

When Hutcheson was grown at 25°C, but then exposed to a higher temperature of 30°C just before and at the time of VPD measurements, the TR response remained linear (Fig. 1C). However, when Hutcheson was grown at 30°C and then measurements were made at 35°C, instead of leveling off at a VPD above 1.9 kPa, TR continued to increase (Fig. 1D). The TR response over the entire range of VPD was linear, which indicated weak



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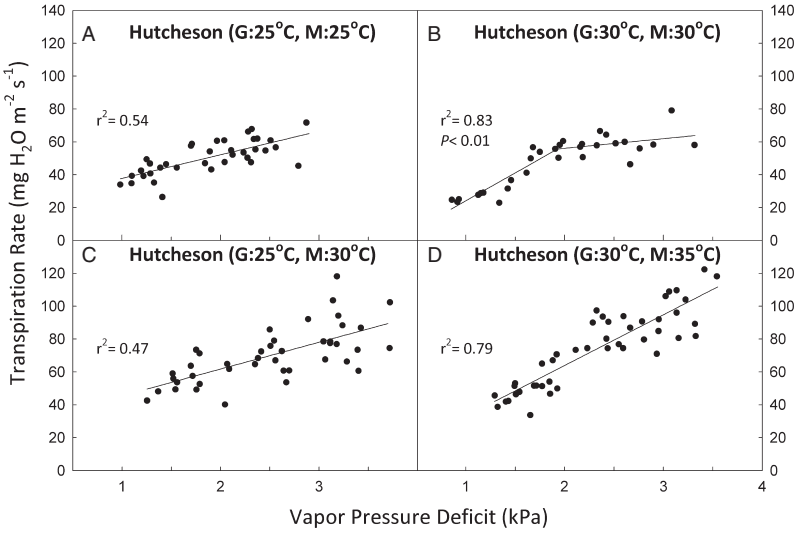



**Fig. 1.** TR (expressed per unit leaf area) response to VPD for *Glycine max* genotype Hutcheson. Growth temperature denoted by 'G' and measurement temperature denoted by 'M' in parentheses in panel title. Best-fit regressions are shown with corresponding $r^2$ values. *P* values are presented only in cases where the two-segment model was the best fit ($P < 0.05$). Regression coefficients can be found in Tables 3 and 4.

**Table 3.** Results of regression analysis of TR on VPD. The best-fit regression and corresponding regression coefficients, $S_1$, $S_2$ and $X_0$, ±SE are reported for two *Glycine max* and four *Glycine soja* genotypes grown at 25°C. Plants were measured for one day at the growth temperature (25°C) and again the next day immediately after the temperature was increased 5°C (25°C→30°C). $S_1$ is the slope of a linear regression, or the slope of the segment before the breakpoint of a two-segment regression; $S_2$, the slope of the segment after the breakpoint of a two-segment regression; $X_0$, the breakpoint of a two-segment regression.

| | *Glycine max* | | *Glycine soja* | | | |
|---|---|---|---|---|---|---|
| | Hutcheson | PI 416937 | PI 468917 | PI 597454A | PI 468397A | PI 549033 |
| | 25°C | | | | | |
| Best-fit Regression | Linear | Linear | Two-segment | Two-segment | Two-segment | Linear |
| $S_1$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | 14.4 ± 2.3 | 14.9 ± 2.4 | 40.8 ± 5.5 | 35.1 ± 0.0 | 45.7 ± 6.4 | 28.5 ± 2.4 |
| $S_2$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | – | – | 10.9 ± 2.7 | −0.53 ± 0.0 | 10.6 ± 2.6 | – |
| $X_0$, kPa | – | – | 1.16 ± 0.1 | 1.45 ± 0.0 | 1.14 ± 0.1 | – |
| $r^2$ | 0.54 | 0.50 | 0.92 | 0.89 | 0.92 | 0.83 |
| n | 42 | 42 | 30 | 33 | 30 | 30 |
| | 25°C→30°C | | | | | |
| Best-fit Regression | Linear | Linear | Two-segment | Two-segment | Two-segment | Linear |
| $S_1$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | 16.3 ± 2.7 | 18.3 ± 2.4 | 33.7 ± 3.3 | 35.0 ± 3.9 | 43.1 ± 7.2 | 26.3 ± 2.6 |
| $S_2$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | – | – | 4.39 ± 4.5 | 13.1 ± 2.7 | 9.25 ± 4.8 | – |
| $X_0$, kPa | – | – | 2.00 ± 0.1 | 1.81 ± 0.2 | 1.76 ± 0.2 | – |
| $r^2$ | 0.47 | 0.59 | 0.89 | 0.94 | 0.87 | 0.79 |
| n | 42 | 42 | 30 | 33 | 30 | 30 |

stability of the maximum-TR response after a temperature increase.

PI 416937, like Hutcheson, had a linear TR response to increasing VPD at a constant temperature of 25°C (Fig. 2A), and exhibited TR sensitivity to high VPD at a constant temperature of 30°C (Fig. 2B). Although the breakpoint was similar (1.91 kPa), PI 416937 had a higher $S_1$ and a lower $S_2$ when compared with Hutcheson (Table 4).

When temperature was raised above the growth temperature of 25 to 30°C at the measurement time, the TR of PI 416937 continued to be a linear function of VPD (Fig. 2C), like Hutcheson. But, when the 30°C growth temperature was increased to 35°C, PI 416937 retained the ability to limit TR when VPD increased above 2.1 kPa (Fig. 2D). However, the slope of the regression after the breakpoint was noticeably greater after the temperature change ($S_2 = 20.8$ mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$) compared with that



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**Table 4.** Results of regression analysis of TR on VPD. The best-fit regression and corresponding regression coefficients, $S_1$, $S_2$ and $X_0$, ±SE are reported for two *Glycine max* and four *Glycine soja* genotypes grown at 30°C. Plants were measured for one day at the growth temperature (30°C) and again the next day immediately after the temperature was increased 5°C (30°C→35°C). $S_1$ is the slope of a linear regression, or the slope of the segment before the breakpoint of a two-segment regression; $S_2$, the slope of the segment after the breakpoint of a two-segment regression; $X_0$, the breakpoint of a two-segment regression.

| | *Glycine max* | | *Glycine soja* | | | |
|---|---|---|---|---|---|---|
| | Hutcheson | PI 416937 | PI 468917 | PI 597454A | PI 468397A | PI 549033 |
| | 30°C | | | | | |
| Best-fit Regression | Two-segment | Two-segment | Linear | Linear | Linear | Linear |
| $S_1$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | 34.0 ± 5.4 | 42.1 ± 6.2 | 25.8 ± 1.8 | 31.7 ± 1.5 | 29.7 ± 1.6 | 34.8 ± 2.2 |
| $S_2$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | 5.85 ± 4.2 | 1.22 ± 4.0 | – | – | – | – |
| $X_0$, kPa | 1.93 ± 0.2 | 1.91 ± 0.1 | – | – | – | – |
| $r^2$ | 0.83 | 0.80 | 0.92 | 0.97 | 0.95 | 0.92 |
| n | 30 | 36 | 18 | 18 | 21 | 24 |
| | 30°C→35°C | | | | | |
| Best-fit Regression | Linear | Two-segment | Linear | Linear | Linear | Linear |
| $S_1$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | 30.9 ± 2.5 | 51.3 ± 11 | 23.9 ± 4.3 | 32.9 ± 4.2 | 26.7 ± 2.5 | 28.2 ± 2.8 |
| $S_2$, mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ | – | 20.8 ± 3.5 | – | – | – | – |
| $X_0$ , kPa | – | 2.10 ± 0.2 | – | – | – | – |
| $r^2$ | 0.79 | 0.85 | 0.71 | 0.83 | 0.90 | 0.88 |
| n | 45 | 45 | 15 | 15 | 15 | 15 |

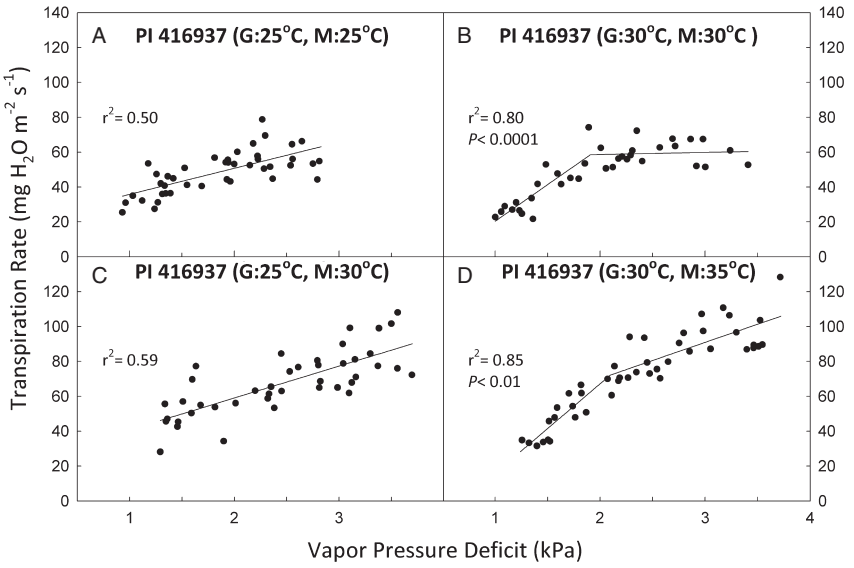



**Fig. 2.** TR (expressed per unit leaf area) response to VPD for *Glycine max* genotype PI 416937. Growth temperature denoted by 'G' and measurement temperature denoted by 'M' in parentheses in panel title. Best-fit regressions are shown with corresponding $r^2$ values. *P* values are presented only in cases where the two-segment model was the best fit ($P < 0.05$). Regression coefficients can be found in Tables 3 and 4.

when the growth and measurements were both at 30°C ($S_2 = 1.22$ mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$) (Table 4).

### *Glycine soja*

The response to VPD at different temperatures of *G. soja* was different than that of *G. max*. Results for the four genotypes are shown in Tables 3 and 4. Two genotypes were used to illustrate the different VPD responses graphically (Figs 3, 4).

The TR response to VPD and its temperature interaction were variable within the *G. soja* accessions. One genotype, PI 549033 (USDA-ARS National Genetic Resources Program, 1991), had a linear response to VPD at a constant temperature of 25°C (Fig. 3A). At that temperature, the three other genotypes were best



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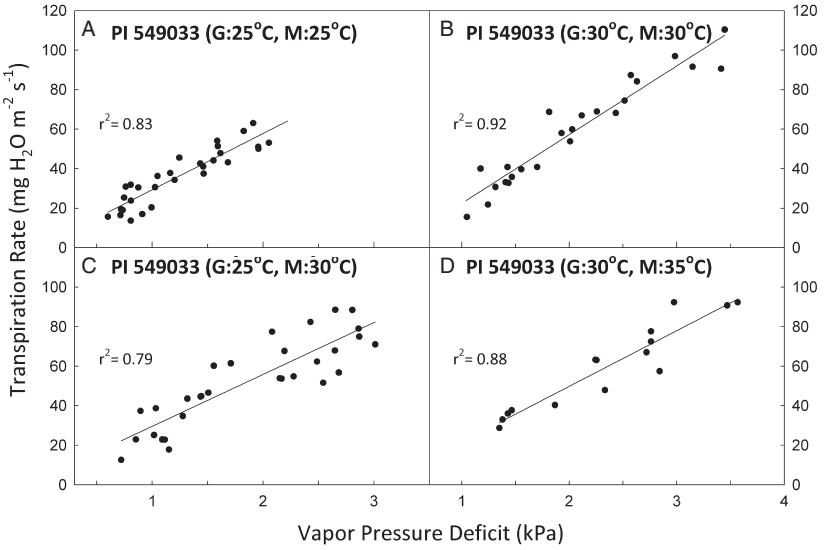



**Fig. 3.** TR (expressed per unit leaf area) response to VPD for *Glycine soja* genotype PI 549033. Growth temperature denoted by 'G' and measurement temperature denoted by 'M' in parentheses in panel title. Best-fit regressions are shown with corresponding $r^2$ values. Regression coefficients can be found in Tables 3 and 4.

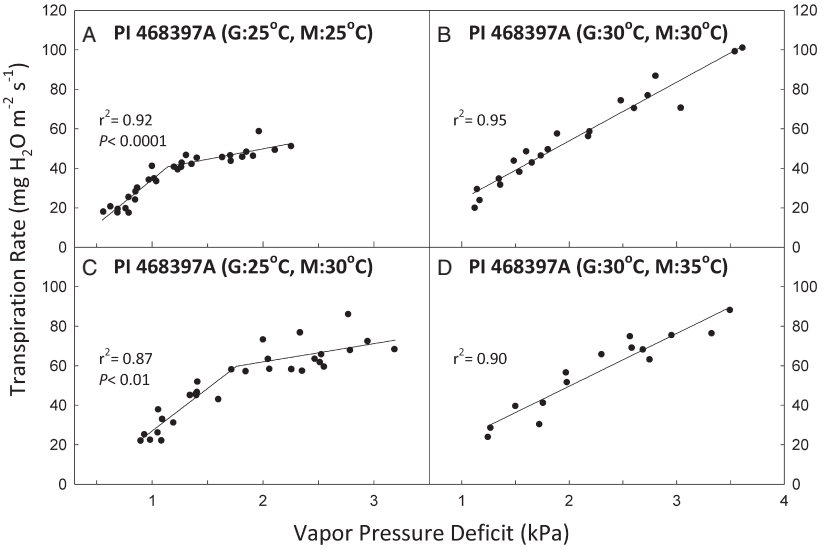



**Fig. 4.** TR (expressed per unit leaf area) response to VPD for *Glycine soja* genotype PI 468397A. Growth temperature denoted by 'G' and measurement temperature denoted by 'M' in parentheses in panel title. Best-fit regressions are shown with corresponding $r^2$ values. *P* values are presented only in cases where the two-segment model was the best fit ($P < 0.05$). Regression coefficients can be found in Tables 3 and 4.

fit by the two-segment regression (Fig. 4A, Table 3). PI 468397A and PI 468917 (USDA-ARS National Genetic Resources Program, 1982a, 1982b) had similar breakpoints of 1.14 and 1.16 kPa, while PI 597454A (USDA-ARS National Genetic Resources Program, 1997) had a higher breakpoint of 1.45 kPa. In contrast to the responses at 25°C, at the 30°C growth and measurement regime, TR of all genotypes was a linear function of VPD (Figs 3B, 4B; Table 4).

The *G. soja* genotypes exhibited the same general TR responses to VPD when the temperature was shifted upward from 25 to 30°C. The genotype that had exhibited a linear response when grown and measured at 25°C continued to respond linearly after the shift to 30°C, while those genotypes with two-segment responses at continuous 25°C continued to exhibit breakpoints after the temperature shift. For the genotypes with breakpoints, some differences were apparent in the



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VPD response profiles. For example, PI 597454A had an $S_2$ value of $-0.53\ mg\ m^{-2}\ s^{-1}\ kPa^{-1}$ when measured at the 25°C growth temperature, but $S_2$ increased to $13.1\ mg\ m^{-2}\ s^{-1}\ kPa^{-1}$ when the measurement temperature was increased to 30°C (Table 3). Therefore, TR at high VPD was effectively controlled at 25°C, but after the temperature increased to 30°C, control was less effective. The other two *G. soja* genotypes with breakpoints had $S_2$ values that were similar to their respective values before and after the temperature shift (Table 3). There was a consistent trend for the breakpoint value to increase with the shift in temperature. For example, PI 468397A had a pronounced shift in its breakpoint from 1.14 to 1.76 kPa (Fig. 4A, C).

When *G. soja* was grown at a constant temperature of 30°C and then switched to 35°C for the VPD measurements, the TR responses were consistent, remaining a linear function of VPD for all *G. soja* accessions (Table 4). While the response pattern was uniform among the genotypes, there were some differences in $S_1$ values.

## Discussion

### Temperature interactions

A main purpose for these experiments was to determine if temperature altered TR responses to VPD. The results clearly indicated that temperature and TR interactions are present. The interaction was prominent in both *G. max* and *G. soja*, although it was expressed differently in the two species and among genotypes. The *G. max* genotypes did not restrict TR in response to VPD at 25°C but, when grown at 30°C, the increase in TR was noticeably limited as VPD increased beyond approximately 1.9 kPa. With an abrupt temperature shift upward from 30 to 35°C, the VPD-sensitive response of the variety Hutcheson was lost, but it was retained to an extent by 'slow wilting' PI 416937.

In most *G. soja* genotypes, a restriction of TR was present as VPDs increased above 1.1–2.0 kPa at the cooler growth temperature of 25°C, but no TR restriction was detected when *G. soja* was grown at the higher temperature of 30°C. A similar sort of temperature sensitivity was observed in recent experiments with maize hybrids, where upward shifts in VPD breakpoints indicated somewhat less effective control of transpiration (Yang et al. 2012).We are aware of no directly comparable results to ours with soybean in the published literature. In one study evaluating $CO_2$ effects on soybean, results from the ambient $CO_2$ treatment did suggest that stomatal conductance decreased as VPD increased, but the sensitivity of this response was diminished by high temperature (Wilson and Bunce 1997).

Differential expression of the VPD-temperature interactions in *G. soja* and *G. max* appears reasonable when the background of these genotypes is considered. Both species are thought to originate from the temperate climate of China (Hymowitz and Singh 1987). While the temperature optimum of *G. soja* is not well documented, the North China Plain, the source of the *G. soja* accessions employed in this study, has average maximum temperatures near 20°C (Mo et al. 2009). Previous studies of soybean and various weed species have suggested that temperature adaptations tend to be a highly conserved physiological trait (Tungate et al. 2007). Therefore, the expression of TR control in *G. soja* in response to increasing VPD only at 25°C reflects the temperate climate adaptation. The *G. max* variety Hutcheson, on the other hand, is a product of breeding targeted to the southern USA climate, where a transpiration-related trait of value would be selected under higher temperature conditions; hence, the ability to limit TR at the higher 30°C growth temperature. Indeed, the temperature optimum of southern USA-adapted soybean genotypes has been reported previously to be near 30°C (Baker et al. 1989, Tungate et al. 2007). Compared with Hutcheson, the exotic genotype PI 416937 displayed slightly tighter control of TR at the 30°C growth and measurement temperature, i.e. a lower $S_2$ slope with increasing VPD above 1.9 kPa; also, its TR control persisted to some extent after the sudden shift to the high 35°C temperature. This TR control by PI 416937 helps to explain why it is a valued source of genetic traits in development of drought resistant soybean germplasm (Carter et al. 2006).

An initial hypothesis was that *G. soja* might serve as a source for drought resistance traits for soybean breeding. The results indicated that genetic diversity for the TR response to VPD is present among *G. soja* accessions. Marked differences were apparent in slopes and breakpoints at 25°C among the genotypes. The breakpoints for TR control were at lower VPD than in soybean, a sensitivity trait that could offer an advantage in water retention. However, because the TR control was limited to the lower temperature range, expression of the trait appears problematic in the development of soybean germplasm suited to warmer areas like those in the southern USA, unless the breakpoint/temperature linkage can be separated.

Temperature dependent VPD responses can explain some of the inconsistencies in the literature for soybean. In experiments with the Monsanto cultivar A5959 (MG V), for example, a two-segment TR response to VPD was observed in one study when the average measurement temperature was 30°C (Sinclair et al. 2008), but a linear TR response to VPD occurred in



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another when the average growth temperature was 34.9°C (Fletcher et al. 2007). The results measured at 30 and 35°C thus were similar to those in our experiments with Hutcheson. The similarities between A5959 and Hutcheson extended further, as $S_1$ and $S_2$ values of 33.2 and 5.7 mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ for A5959 at 30°C (Sinclair et al. 2008) were quite similar to the 34 and 5.8 mg m$^{-2}$ s$^{-1}$ kPa$^{-1}$ observed here with Hutcheson at 30°C (Table 4). And the slope of the linear response for A5959 at 35°C (Fletcher et al. 2007) was similar to that for Hutcheson. We cannot know with certainty if there is a genetic connection between A5959 (a proprietary Monsanto variety) and Hutcheson, but Hutcheson was one of the most-used parents between 1990 and 2008 for the Southern germplasm of Monsanto (Mikel et al. 2010), so a genetic link seems possible. If true, it would show that TR control traits are heritable and available for soybean improvement. A previous greenhouse study focused on variability in a recombinant inbred soybean population revealed complex heritability of the TR-VPD trait (Sadok and Sinclair 2009). Those experiments may have been compromised, however, by differing growth temperatures which ranged from 20 to 33°C. Leaf VPD responses, of course, are just one type of trait that can contribute to drought-stress resistance *in situ*, and field studies indicate there may be some differences between A5959 and Hutcheson. In one comparison, A5959 was observed to be a fast-wilting cultivar while Hutcheson was intermediate (King et al. 2009).

## Mechanistic implications

The TR response profiles of *G. soja* and *G. max* offer insights into the mechanism(s) controlling water release and the intricacies of inherited drought resistance traits in soybean. Two kinetically definable transpiration systems are present. When TR increases were restricted above a VPD of 1.9 kPa in soybean at 30°C (Figs 1B, 2B), for example, the TRs occurring below 1.9 kPa were unaffected. Thus 'System I', TRs occurring below 1.9 kPa, exhibited little or no plant TR control. In contrast, the segment of the TR response curve above 1.9 kPa was tightly constrained in both soybean genotypes. Regulation of the second segment, or 'System II', controlled the upper capacity for water movement through the plant. The VPD responses to temperature indicated that System II is dynamic and influenced by the growth environment. Expression of System II regulation also appears to become engaged in a temperature range where maximal growth rates occur, 30°C for *G. max* and 25°C for *G. soja*.

Recognition of the System II response underscores the importance of 'physiological plasticity' in controlling TR under high VPD. As with all crop plants in the field, soybean experiences short term changes in VPD during a day as atmospheric vapor pressure changes. In summer months, the VPD changes most often would be accompanied (and driven) by increasing temperatures as a day progresses. The absence of a definable breakpoint with Hutcheson when it was shifted to 35°C implies that regulation of System II became inoperable. The presence of a two-phase response to VPD by soybean PI 416937 at 35°C indicates that some degree of System II regulation persisted in this slow-wilting genotype. All three *G. soja* genotypes exhibiting the System II response pattern retained System II regulation with the shift to the higher temperature above its 25°C growth optimum, but $S_2$ values tended to vary (Table 3). Thus, genetic diversity for the 'plasticity' of the System II regulatory response was present in the wild progenitor to soybean when it was expressed. The exotic soybean PI 416937 obviously has retained elements of the System II regulatory trait. It was observed previously that PI 416937 exhibited stress responses that were intermediate between native weed species and modern soybean cultivars (Place et al. 2008).

Rapid changes in the engagement of System II in the soybean species indicate that the capacity for water movement through the plant probably is not because of a structural limitation (Sinclair et al. 2008). If the xylem carrying capacity were limiting, for example, then one might expect TR responses to VPD to remain stable when plants were exposed to short-term temperature changes. The similarity of $S_2$ values in one of the *G. soja* genotypes (PI 468397A) before and after the temperature shift from 25 to 30°C means the possibility of structural limitations cannot be dismissed in that instance. So what internal mechanism regulates System II in *G. max and G. soja* with temperature-sensitive $S_2$ values? While the results of this study do not offer direct evidence to resolve this question, recent experiments with soybean PI 416937 indicated hydraulic conductance between the xylem and guard cells of PI 416937 may be controlled by an aquaporin (AQP) family sensitive to AQP inhibitor $AgNO_3$ (Sadok and Sinclair 2010a, 2010b), and AQP function seems to be temperature sensitive (Cochard et al. 2007, Ionenko et al. 2010). If AQPs were involved, it opens the possibility that AQP-specific molecular markers could be used to screen soybean populations for TR regulation at higher VPDs.

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